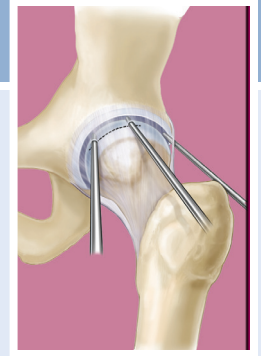


Hamstring Injuries

Kyle E. Hammond, Lee M. Kneer



INTRODUCTION

Hamstring injuries are common in athletic populations and can affect athletes at all levels of competition.¹⁻⁴ Several studies have shown that the rates of muscle strain in high school football (12% to 24%) and collegiate football (18.9% to 22.2%) are fairly high.⁵⁻⁹ In one study, injury surveillance conducted by the National Football League identified 1716 hamstring strains among all players, with a range of 132 to 210 injuries per year, which accounts for an overall injury rate of 0.77 per 1000 athlete-exposures and a reinjury rate of 16.5%.¹⁰ Hamstring injuries accounted for nearly one-fifth of all injuries sustained by elite track and field athletes, with males at twice the risk compared with their female counterparts.¹¹ One literature review identified previous hamstring injury as the greatest risk factor for reinjury, with age and male sex also increasing injury risk.¹² The injured muscle may have an altered compliance or deformation pattern, predisposing it to reduced tensility or higher muscle strain. Although some studies suggest that contact activities are the cause of hamstring injuries, most studies have shown that more than 90% of injuries occur without contact, with the classic injury being sustained by a water skier who gets pulled up by the boat, resulting in abrupt knee extension and hip flexion.^{5,13}

The hamstring complex consists of the short and long heads of the biceps femoris, semitendinosus, and semimembranosus. The semitendinosus, semimembranosus, and long head of the biceps are biarticular and are innervated by the tibial portion of the sciatic nerve. The short head of the biceps is monarticular and innervated by the common peroneal nerve. These muscles work together to extend the hip, flex the knee, and externally rotate the hip and knee; they have significant overlap at the myotendinous junction.¹²

The proximal hamstring complex has a strong bony attachment on the ischial tuberosity (Fig. 86.1).¹⁴ The ischial footprint is composed of the common proximal tendon of the semitendinosus and the long head of the biceps femoris as well as a distinct semimembranosus footprint. The semimembranosus passes anteriorly to the conjoined semitendinosus/biceps femoris tendon to its origin on the lateral ischium (see Fig. 86.1).

Biomechanically the hamstrings are subjected to high tensile load, given their extensive eccentric role. Hip and knee flexion during the initial swing phase requires simultaneous eccentric and concentric activity of the hamstrings. During terminal swing,

the hamstrings continue to play a dual role of preventing knee hyperextension while opposing hip flexion.¹⁵ The hamstrings work synergistically with the gluteal muscles to stabilize, decelerate, and propel the hip. During the propulsion phase, the medial hamstrings assist in decelerating hip external rotation, which maintains the gluteus maximus at an ideal length to act as an accelerator (along with the hamstrings) of the femur in the sagittal plane. The hamstrings along with the rectus abdominis are also decelerators of pelvic anterior tilt throughout stance. Given these functional relationships, it is conceivable that hamstring strain or rupture has its source in the inhibition and weakness of its closest synergists, the gluteal and abdominal muscles; these muscle groups have become targets for injury prevention and rehabilitation.¹⁶

Hamstring injuries most commonly occur proximally, as one study of 275 soccer players has demonstrated.¹⁷ In this study of hamstring strains, purely tendinous injuries (7.9% of all hamstring strains) were relatively rare, with the proximal myotendinous junction, muscle belly, and distal myotendinous junction (36.1%, 32%, and 17.7%, respectively) implicated more frequently. The long head of the biceps femoris was most commonly (56.5%) implicated, whereas injuries to the short head of biceps femoris occurred only 5.6% of the time.¹⁷

Hamstring injuries occur on a continuum that can range from musculotendinous strains to avulsion injuries.^{1,2} A strain is a partial or complete disruption of the musculotendinous unit.^{1,4} A complete tear or avulsion, in contrast, is a discontinuity of the unit. Distal avulsion injuries are quite rare and not discussed in this chapter. Most hamstring strains do not require surgical intervention and resolve with a variety of modalities and rest. Differentiating the complete and partial tears from the muscle strain subgroup is of importance, as patients with complete or partial tears can experience more substantial disability.

HISTORY

The history of an acute injury classically involves a traumatic event with forced hip flexion and the knee in extension, as observed in water skiing, although studies have suggested that most injuries occur during high-speed running or moments of rapid change of pace.^{2,18-23}

Commonly athletes with proximal hamstring tendon tears describe an acute popping or tearing sensation with associated function-limiting pain.^{24,25} Although pain is typically the

Abstract

Hamstring injuries are common at all levels of athletics. They occur most often within the muscle belly or musculotendinous regions; however, they can also involve the proximal and/or distal attachments. Muscle injuries are typically managed by nonoperative means and are associated with a high rate of return to activity and full recovery. Nonoperative methods include rest, physical therapy, medications, and sometimes injection of either a corticosteroid or a biologic. Proximal injuries can also be strains that will heal through nonoperative means, but these injuries can also be traumatic ruptures with retraction of the tendon or chronic injuries that require surgical repair to heal properly and return the patient to his or her desired activity levels. Surgical repair is recommended to maximize outcomes when there is a full rupture of at least two tendons at a retracted distance of at least 2 cm in a healthy individual. The repair can be performed in an open or endoscopic manner. The postoperative recovery can take 6 to 9 months to fully return the athlete to sport. Complications with both nonoperative and operative repairs exist. Sciatic nerve injuries can occur in both scenarios. Surgical cases typically have a higher risk of complication due to wound issues, infections, nerve injuries, sitting pain, and residual weakness. Overall, outcomes for both nonoperative and operative care of hamstring injuries result in a high percentage of good to excellent outcomes and return to activity.

Keywords

hamstring strain
proximal hamstring rupture
physical therapy
platelet-rich plasma
surgical repair
sciatic nerve

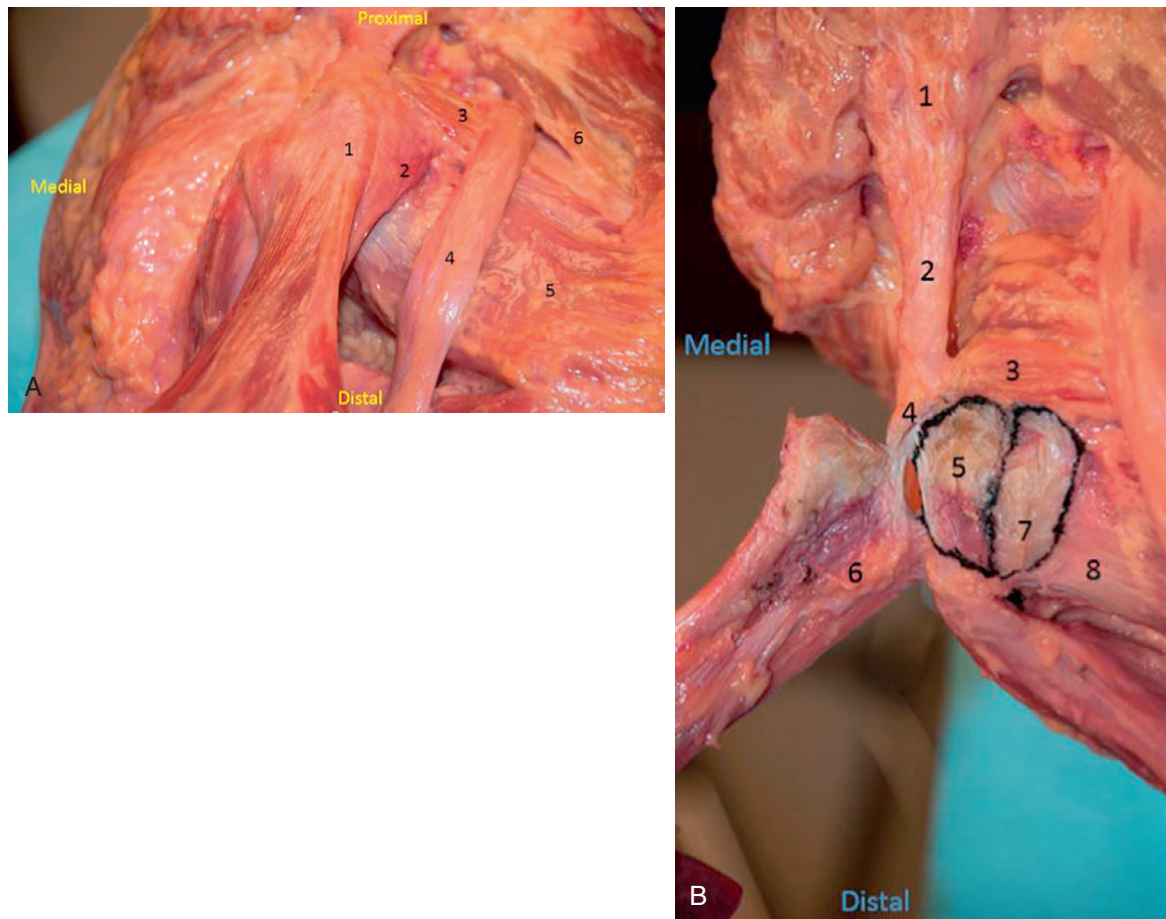


Fig. 86.1 Normal anatomy of the hamstring origin. (A) The posterior view of a cadaveric dissection of the ischium in a left hip. The conjoined semitendinosus/biceps femoris tendon (1) passes posteriorly to the semimembranosus tendon (2) to insert medially on the proximal hamstring footprint at the posterolateral ischium. The inferior gemellus (3), sciatic nerve (4), quadratus femoris (5), and piriformis (6) are closely associated with the proximal hamstring. (B) The footprint of the hamstring on the posterolateral ischium is composed of the conjoined tendon (5) medially and the semimembranosus (7) laterally. Structures near the hamstring footprint include the sacrum (1), sacrotuberous ligament (2), inferior gemellus (3), medial bony prominence (4), conjoined tendon (6), and quadratus femoris (8).

chief complaint, the patient may also complain of a sense of instability or gait incoordination.^{22,25-28} Occasionally patients who present with hamstring tears may report altered sensation or pain in a dermatomal distribution much like that seen in radiculopathy.^{22,25,27,29} This may be explained by acute nerve traction, the caustic effect of blood on the nerve, or compression from the associated hematoma. The patient may complain of a previous similar injury in the same leg, given the high risk of recurrence.³⁰

PHYSICAL EXAMINATION

The examination is typically performed with the patient in the prone position. Maintaining the knee in a slightly flexed position will make the examination more comfortable for individuals with more severe injuries. Inspection and palpation of the posterior thigh may reveal fasciculations or muscle spasm. Ecchymosis may be observed if the fascial covering is also disrupted. Palpation of the entire posterior thigh is important to localize the

injury. Palpation should be performed systematically, starting at the broad insertion along the ischial tuberosity. Identification of a more proximal injury is important, as studies have shown a prolonged recovery course for injuries occurring more proximally, especially within the tendon.^{31,32} Given the biarticular nature of the hamstring complex, it is important to examine the strength of both knee flexion and hip extension as well as range of motion (ROM). The authors suggest evaluating knee flexion in both relative extension (15 to 30 degrees flexion) and flexion at 90 degrees as well as hip extension with the knee fully extended and again at 90 degrees of knee flexion while the patient is prone. Eccentric testing of the hamstrings via resisted extension of the knee from 90 to 15 degrees may elicit pain with milder injuries. Careful note should be taken of side-to-side strength differences. Similarly, a decrement in painless ROM may be used to identify a more chronic hamstring injury, and several examination maneuvers have been described to assess for tendinopathic functional changes.³³ Our preferred test for proximal tendinopathy pain is a supine single-leg plank.

During examination, one must have a high index of suspicion for a tear. In less acute situations where the tear is several days old, it is possible that even a large defect may not be palpable clinically owing to the overlying hematoma. It is especially critical to assess these patients with imaging studies to delineate the type of tear that is present and to guide management.

IMAGING

No initial imaging is indicated for hamstring injuries involving no loss of strength and minimal to moderate pain, especially if discomfort is isolated to the muscle belly. If pain is more severe and located proximally, plain radiographs including an anteroposterior view of the pelvis and lateral image of the affected hip are warranted to rule out and characterize an avulsion injury. If a fracture is identified, computed tomography (CT) or magnetic resonance imaging (MRI) may assist in assessing the displacement and fracture configuration for possible surgical planning. Special consideration should be given to obtaining radiographs in the younger patient at risk for apophyseal injuries (Fig. 86.2).

MRI and ultrasound are the imaging modalities of choice to evaluate nonavulsion injuries resulting in weakness, gait abnormalities, or severe pain or in lesser injuries involving elite athletes.³⁴ MRI has been shown to be more accurate in evaluating tears, especially to deeper tissues or in patients with recurrent tears, as scar tissue can be more easily mistaken for acute injury with ultrasound. However, studies have not examined this in recent years, as ultrasonographic image resolution has greatly improved.³⁵ MRI findings, particularly the Peetrons classification and size of muscular edema, have been shown to be predictive of return to play (RTP) following grade 1 and 2 injuries.^{36,37} Data support the use of MRI as the preferred tool for the evaluation of tendon retraction in the case of a complete tear as well, making it particularly helpful in surgical planning (Fig. 86.3). Incomplete tears without retraction often reveal a “sickle sign,” as demonstrated by a curvilinear signal on T2-weighted images (Fig. 86.4).²³ One must be cautious in considering advanced

imaging, as partial hamstring tears have been detected in up to 15% of asymptomatic patients.^{38,39}

Ultrasound is useful as a point-of-care examination tool; it allows for dynamic imaging and comparison to the contralateral limb; it is also far less expensive than MRI. Recent advances in technology have allowed for the development of portable units with resolution allowing for their use as sideline tools, and there are no considerations for patient mass, implanted devices, or claustrophobia. Ultrasound is highly user-dependent and not available universally. Moreover, the evaluation of pathology can

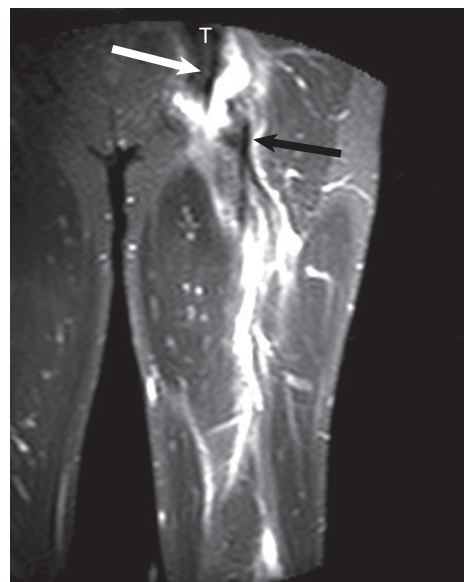


Fig. 86.3 A coronal T2-weighted magnetic resonance image of a complete three-tendon rupture of the proximal hamstring. The *black arrow* points to the common avulsed tendon. The *white arrow* identifies the tuberosity (T).



Fig. 86.2 An anteroposterior radiograph of the pelvis showing a bony avulsion of the right ischial tuberosity.

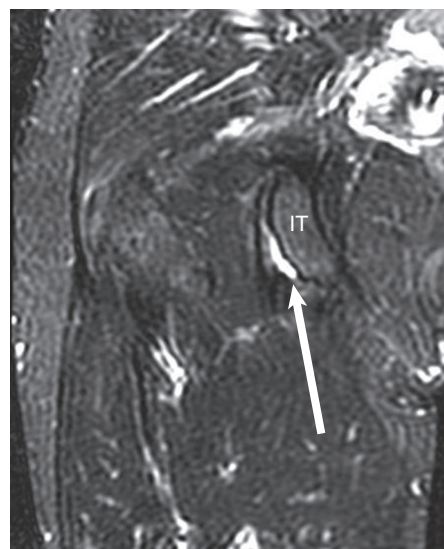


Fig. 86.4 A coronal T2-weighted magnetic resonance image of a right hip showing the sickle sign (*arrow*), which indicates fluid within the ischial bursa. IT, Ischial tuberosity.

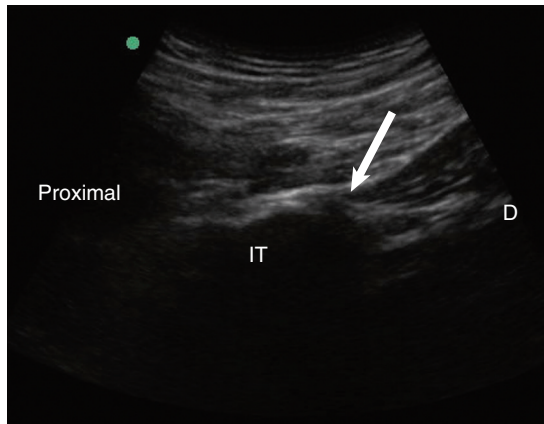


Fig. 86.5 Ultrasound image of the origin of the left hamstring (arrow). D, distal; IT, ischial tuberosity.

be hindered by the presence of large hematomas, as is often common in hamstring tears (Fig. 86.5). Despite this, ultrasound has been shown to be highly accurate in the evaluation of partial tears and insertional tendinosis.⁴⁰

DECISION-MAKING PRINCIPLES

Whether the surgical procedure is performed with an open approach or endoscopically, the indications are the same. The only certain indication for the open procedure is a large retracted tear with chronic atrophy as noted on MRI. In these cases, the procedure would more than likely require extensive mobilization and probably the use of a graft for reconstruction of the avulsed segment, which would have to be performed in an open fashion at this time, usually with a longitudinal incision. The first indication for surgery is an acute hamstring avulsion of at least two out of the three tendons in an active patient with greater than 2 cm of retraction. Some patients have a clinically evident partial hamstring avulsion involving the biceps/semitendinosus tendon, with refractory ischial pain and the inability to return to high-level sports. Finally, patients are also candidates for surgical intervention when they have a history of refractory ischial bursitis and no discernible tear and conservative treatment has failed, including at least 12 weeks of physical therapy and potentially ultrasound-guided ischial injections with corticosteroids and/or biologics.

TREATMENT OPTIONS

Nonoperative

Initial nonoperative treatment is warranted for the majority of hamstring injuries, with surgical management reserved for larger avulsions or large muscular tears. Tenets include active rest, the use of oral nonsteroidal antiinflammatory medications, and a physical therapy program consisting of gentle hamstring stretching and strengthening. As the initial inflammation resolves, core, hip, and quadriceps exercises can be added in association with a more aggressive hamstring injury prevention program.^{41,42} In a recent randomized controlled trial, male soccer players showed a decrease in hamstring injuries with the implementation

of Nordic hamstring exercises as part of the standard conditioning program.⁴³ Progression then advances to sport-specific exercises, and full participation is allowed when the patient can perform these without symptoms.¹² If progress does not occur with this program, an ultrasound-guided corticosteroid injection may be used; this has been shown to be both safe and effective at providing significant relief and allowing a hastened (RTP).⁴⁰ Fader et al. showed that in patients with refractory chronic proximal hamstring tendinopathy, platelet-rich plasma injections can improve pain scores compared with other nonoperative means.⁴⁴ Patients who experience failure of nonoperative treatment for partial tears may benefit from surgical débridement and repair, similar to patients with other commonly seen partial tendon tears (i.e., the rotator cuff). As described further on, newer and less invasive endoscopic techniques are also an option for this problem.

Nonoperative treatment of complete ruptures of the proximal hamstring or those with more than 2 cm of retraction is less frequently recommended because surgical repair has resulted in the successful return of patients to a high level of function.^{18,20,24,25,29,45-49} One study showed that up to 40% of patients with partial proximal hamstring tears and less than 2 cm of retraction go on to require surgical intervention, and other studies have shown similar results.^{20,46} Partial hamstring injuries have been shown to be successfully treated with débridement and suture anchor placement after a period of failed conservative care; however, this approach is less effective in the treatment of complete tears or those treated acutely.⁴⁷

Surgical Treatment

Open

Open techniques are still the standard of care and are most often described in the literature. The indications for surgical treatment of proximal hamstring ruptures include all acute complete three-tendon tears and two-tendon tears with retraction of 2 cm or more.⁴⁸ Acute surgical repair initially is not indicated for patients with a one- or two-tendon tear with less than 2 cm of retraction; they are treated surgically if nonoperative treatment is not successful. In addition, less active patients or patients who are unable to comply with the postoperative rehabilitation protocol should be managed nonoperatively. Patients with complete or partial tears for whom conservative management fails may be candidates for delayed repair.

A transverse incision in the gluteal crease inferior to the ischial tuberosity is used.^{19,21,48,49} The tendon should be placed on the lateral aspect of the ischial tuberosity and should lie down flat to allow optimal bone healing as well as to prevent prominence (Fig. 86.6).

Endoscopic

Some surgeons have begun to utilize an arthroscopic approach.⁵⁰ With this technique, the patient is placed in the prone position after induction of anesthesia, with all prominences and neurovascular structures protected. The table is flat (as opposed to the slightly flexed position of the table in the open repair procedure) to help maintain the space between the gluteal musculature and the ischium.

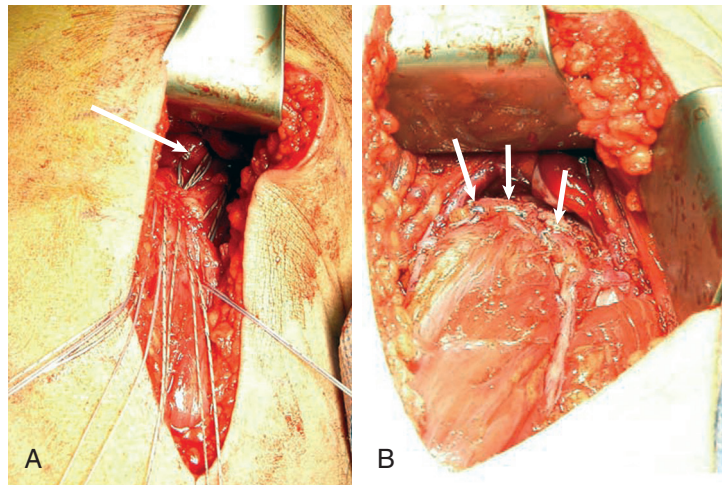


Fig. 86.6 Open repair of the proximal hamstring on the ischial tuberosity (*arrow*). (A) The view is of a longitudinal incision in the left hip. Multiple sutures are in place, with the ischial tuberosity visualized in the wound. (B) Final repair of the hamstring, with multiple suture points visualized (*arrows*).

Two portals are then created, one each 2 cm medial and lateral to the palpable ischial tuberosity (Fig. 86.7). The lateral portal is established first by using blunt dissection with a switching stick as the gluteus maximus muscle is penetrated and the sub-muscular plane is created. The medial portal is then established, taking care to palpate the medial aspect of the ischium. A 30-degree arthroscope is then inserted in the lateral portal and an electrocautery device is placed in the medial portal. The space between the ischium and the gluteus muscle is then developed, taking care to stay along the central and medial portions of the ischium to avoid any damage to the sciatic nerve. With the lateral aspect identified, the dissection continues anteriorly and laterally toward the known area of the sciatic nerve (Fig. 86.8). Very careful and methodical release of any soft tissue bands is then undertaken in a proximal-to-distal direction to mobilize the nerve and protect it throughout the exposure and ultimate repair of the hamstring tendon.

Once the nerve has been identified and protected, attention is directed once again to the area of the tendinous avulsion. The tip of the ischium is identified through palpation with the instruments. The tendinous origin is then inspected to identify any obvious tearing. With acute tears the area is obvious and the tendon is often retracted distally. In these cases, a large hematoma is occasionally present and must be evacuated. It is especially important to protect the sciatic nerve during this portion of the procedure because it is sometimes obscured by the hematoma.

Once the area of pathology has been identified (in persons with incomplete tears), an endoscopic knife can be used to longitudinally split the tendon along its fibers. The hamstring is then undermined and the partial tearing is débrided with an oscillating shaver. The lateral wall of the ischium is cleared of devitalized tissue and a bleeding bed is established in preparation for the tendon repair.

An inferior portal may then be created approximately 4 cm distal to the tip of the ischium and equidistant from the medial and lateral portals. This portal is used for insertion of suture

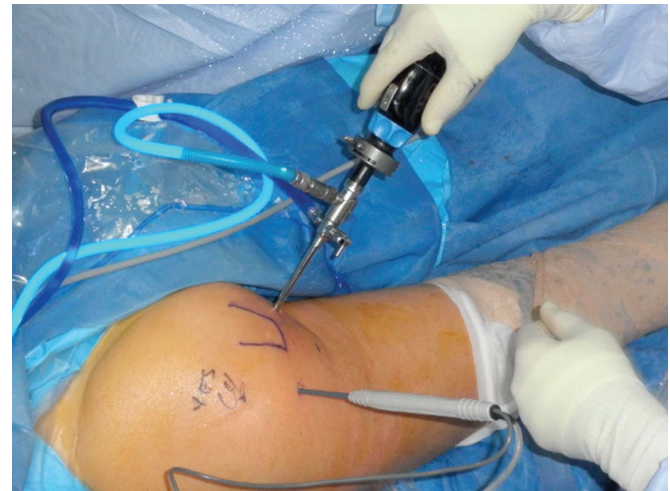


Fig. 86.7 Portals for the endoscopic approach. Note that the arthroscope is in the medial portal and the empty portal is the distal portal. The ablator is in the lateral portal.

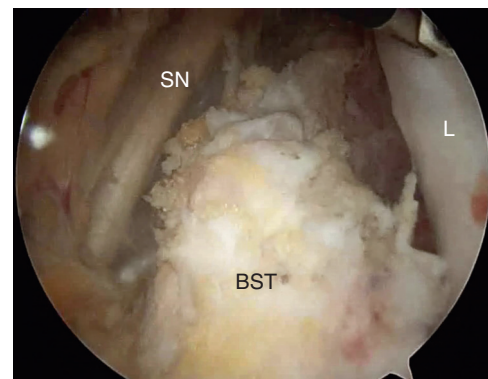


Fig. 86.8 Normal arthroscopic anatomy exposure in a left hip as viewed from the lateral portal. Note the tool entering from the medial portal. BST, common biceps/semitendinosus; L, lateral ischium; SN, sciatic nerve.

anchors as well as suture management. The principles are essentially the same as those used in arthroscopic rotator cuff repair. Once all of the sutures have been passed through the tissue of the avulsed hamstring, the sutures are tied and a solid repair of the tendon is completed.

POSTOPERATIVE MANAGEMENT

Rehabilitation

Assuming that hamstring strain or rupture has its source in the inhibition and weakness of its closest synergists, the gluteal and abdominal muscles, it is important to focus on both groups in the rehabilitation of these injuries, whether they are treated surgically or not. Because the gluteal muscles work in conjunction with the hamstrings to extend the hip, when the agonists (gluteal muscles) for hip extension are weak, an increased relative effort of the hamstrings is required to control trunk and hip flexion

during the loading phase of running.⁵¹ In one prospective evaluation, 24 athletes underwent rehabilitation in one of two protocols: either an isolated hamstring stretching and progressive strengthening protocol or a progressive agility and trunk stabilization protocol.⁴² Follow-up at 2 weeks and 1 year revealed a significantly higher reinjury rate in athletes treated with the isolated hamstring rehabilitation protocol (54.5% vs. 0% and 70% vs. 7.7%, respectively). This finding suggests that strong neuromuscular control of the lumbopelvic region allows the lower extremity muscle to function at high velocity while maintaining a protected ROM for hamstring muscles.

Using the synergistic concept as the premise for the rehabilitation program, the patient ambulates non-weight-bearing on crutches with for the first 4 weeks, depending on the quality of tension on the repair. Weight bearing is advanced to full weight bearing by 6 weeks with continued use of crutches as needed until 8 weeks, as the knee brace is extended to neutral by 6 weeks.



Authors' Preferred Technique

The authors' preferred technique is to perform an outpatient open repair between 2 and 4 weeks postinjury in acute ruptures. With chronic full and partial ruptures that undergo surgical repair, there is no time constraint. Although primary repairs of acute ruptures can be successfully made up to 8 weeks postinjury, it is certainly easier to mobilize the tissue within 4 weeks of the injury. We like to pay close attention to their sciatic nerve examination, both preoperatively and postoperatively. We routinely decompress the sciatic nerve and perform a neurolysis in all cases to optimize the outcome for the nerve. We use a standard prone position with a table break at the patient's iliac crest, which will be flexed about 30 degrees to improve access at the gluteal crease (Fig. 86.9). We place pillows and pads underneath the knee down to the foot, so the knee is in flexed position of about 30 degrees. We utilize a gluteal crease incision for cosmesis and keep this incision as short as safely possible and centered over the lateral aspect of the ischial tuberosity (Fig. 86.10). We utilize a standard fascial layer approach, localize and protect the gluteal neurovascular structures, and then localize the sciatic nerve to perform a neurolysis and protect it throughout the remainder of the case (Fig. 86.11). We then decompress any seroma and/or hematoma and decorticate the footprint of the ischial tuberosity with a rongeur and curette. We then utilize five 3-0 biocomposite, single-loaded SutureTak anchors (Arthrex) in a clustered pattern to fill the footprint. Once this is completed, we begin to suture the tendon after it has been débrided and prepared using a locking stitch and single-pass stitch in a mattress fashion from each anchor. Once all anchors have been sutured, we tie down beginning with the proximal footprint and bringing the knee into 45 degrees of flexion. A layered closure with Vicryl is performed of the hamstring and gluteal fascia; then Monocryl and Dermabond are used for the skin closure. Postoperatively we use a standard hinged postoperative knee brace locked at 30 degrees and make the patient non-weight-bearing for 6 weeks. Partial chronic tears that require surgery are performed in the same manner, and chronic retracted tears are repaired utilizing an Achilles allograft with a longitudinal incision (Fig. 86.12). In the chronic technique, we will place a small bone block into the ischial tuberosity with a biocomposite interference screw and then add additional fixation around the bone block with suture anchors and soft-tissue fixation. We then attach the allograft to the native musculotendinous end of the hamstrings under tension and 90 degrees of knee flexion. Postoperatively, the knee is kept at 90 degrees for 6 weeks in the chronic repairs.

The postoperative protocol is similar for both arthroscopic and open techniques with similar expected outcomes. To date, there is no study to our knowledge that compares outcomes for arthroscopic versus open techniques.



Fig. 86.9 The patient is positioned in the prone position with the break in the operative table flexed to 30 degrees to expose the gluteal fold.



Fig. 86.10 The standard transverse incision is kept as small as possible to perform the surgery safely and adequately, as well as being placed directly in line with the subgluteal skin crease to maximize cosmesis.

Authors' Preferred Technique—cont'd

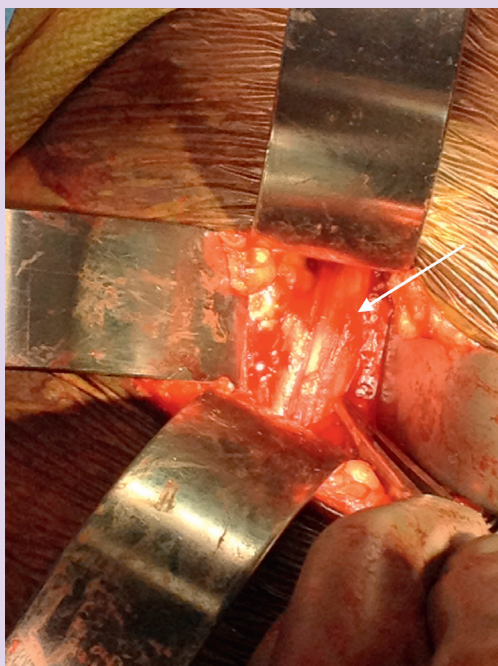


Fig. 86.11 The white arrow points to a sciatic nerve exposed and decompressed in a case of hamstring syndrome.

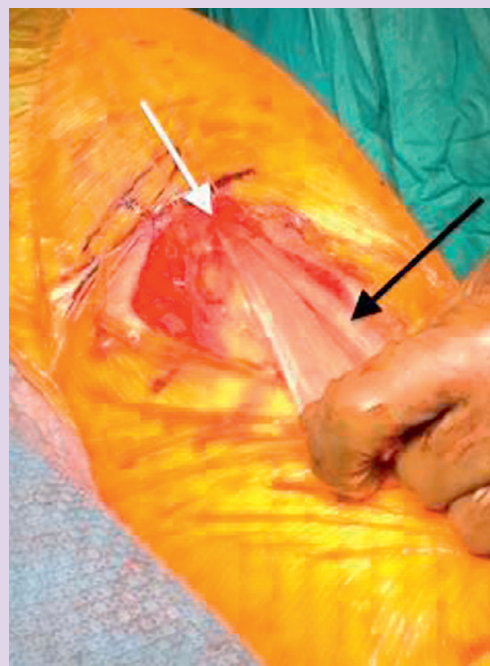


Fig. 86.12 The white arrow points to the ischial tuberosity, where the Achilles allograft (black arrow) has been fixed with an interference screw and suture anchor fixation.

The brace is removed around 8 weeks after surgery. Passive hip ROM is begun with a therapist at 2 weeks and active hip flexion is started at 4 weeks. Isotonic strengthening (at 6 weeks), isokinetic strengthening (8 weeks), and aqua therapy are initiated with progression of core pelvic and closed-chain exercises. Dry land training and sport-specific training are initiated at 12 weeks, with return to full sports participation 6 to 9 months after surgery.

RESULTS

In one study of 52 patients in which the open technique was used for both acute⁵² and chronic¹² repairs, a 96% satisfaction rate was found with the use of subjective validated outcome scales and at an average of 27 months follow-up.⁵³ Lower Extremity Functional Scale results indicated that acute repairs had statistically significant greater outcomes than did chronic repairs ($P = .023$). Thirty-five patients (67%) reported that they could participate in strenuous activities at their latest follow-up. Mansour et al. showed that in the National Football League there was an initial 90% return to sport rate for one game after acute repair of a complete rupture, but only a 50% return to sport rate at the final follow-up; however, the sample size was small.⁵⁴

Another study of eight patients with a similar suture anchor repair revealed postoperative Cybex testing at 88% of peak torque, with the ratio of hamstring-to-quadriceps strength being 0.55, which was not significantly different from the other side.²⁶ An analysis of 10 athletes with acute complete hamstring tendon

tears revealed that the average peak torque was 82% and the hamstring-to-quadriceps strength ratio was 0.56.²⁷ Nine of ten patients returned to their previous level of professional sports activities. Three patients had acute sciatic nerve symptoms that were successfully treated with hematoma evacuation and neurolysis.

Most studies evaluating late surgical repairs have been in patients with chronic tears for complaints of pain while sitting, hamstring weakness, and sciatic nerve symptoms due to scarring, also known as hamstring syndrome.^{24,48,55} These chronic tear repairs yield less consistent results, and the potential exists of scarring of the hamstring-avulsed tissue to the sciatic nerve, which necessitates a sciatic neurolysis. As a result of the inferior results, most surgeons experienced in proximal hamstring repair recommend early reattachment compared with delayed repair.⁵⁶ In the study by Sarimo et al., 41 athletes underwent surgical repair of either acute or chronic proximal hamstring injuries. The authors found that the odds ratio of having a moderate or poor result was 29-fold with a delay of surgical repair of greater than 3 months.²⁵ On the other hand, a more recent study comparing the acute versus chronic repair of complete tears showed that there were good functional and subjective results in both groups with no statistical difference in their outcome scores.⁵⁷ Subbu et al. showed a quicker return to sport in the acutely (within 6 weeks of injury) repaired group between 9 and 13 weeks, as well as fewer postoperative complications.⁵⁸ A recent systematic analysis showed both operative repair and nonoperative management to result in similar

positive outcomes without a significant benefit in surgical repair.⁵⁹ However, as in most systematic reviews, there was heterogeneity of the included studies, thus limiting the ability to utilize these findings clinically.

For patients with chronic ruptures who undergo repair, an allograft may be necessary for reattachment to the ischial tuberosity.⁵⁶ The literature includes few results with regard to this procedure; however, Folsom and Larson⁵² have reported on five patients who required Achilles tendon allograft reconstruction for repair of a chronic rupture. They found that reconstruction of chronic ruptures with an Achilles allograft appeared to restore function and strength at a level comparable with acute repairs.

For patients with high-grade partial insertional tears who fail to respond to nonoperative treatment, surgical repair is performed. The surgical approach is the same as previously described. Once the tendon is exposed, it is incised and released from the tuberosity with use of an elevator. It is then repaired using the same technique for complete tears with suture anchors. Treatment of partial tears has been reported by Lempainen et al., with 41 of 47 athletes returning to sport in average of 5 months.⁶⁰

Chronic proximal hamstring tendinopathy with sciatic nerve symptoms has also traditionally been included in the generic term *hamstring syndrome*. In this specific injury, the tendon is traumatized from repetitive overuse injury. Theoretically the tendon undergoes repetitive stretch and mechanical overload and is unable to fully heal. The sciatic nerve can undergo similar types of stress, leading to scarring, adhesion, and impingement from the thickened tendon. In the cohort in the study by Lempainen et al.,⁶⁰ surgical treatment was performed with tenotomy of the thickened semimembranosus tendon and tenodesis to the biceps femoris with sciatic nerve release; 89% of the patients had good to excellent results.

COMPLICATIONS

Potential complications associated with proximal hamstring ruptures prior to surgical treatment are related to the mechanism of injury. The complications can be early or delayed. The early complications most commonly involve a neurapraxia injury to the sciatic nerve as a result of a stretch injury. Depending on the mechanism and force of the injury, the sciatic nerve can be damaged, leading to burning symptoms radiating down the leg and weakness of the foot. During the initial examination it is critical to determine if the sciatic nerve is functioning appropriately to document and ensure that no iatrogenic injury is present at the time of surgery. Fortunately neurapraxia injury most commonly resolves over time despite being troubling to the patient initially. Delayed complications of nonoperative treatment of proximal hamstring ruptures have been described by Puranen and Orava.⁵⁵ These complications include knee flexion and hip extension weakness, difficulty sitting, hamstring deformity, and the potential development of symptoms similar to those of hamstring syndrome as the tendons scar down to the sciatic nerve. Hamstring syndrome consists of local posterior buttock pain and discomfort over the ischial tuberosity. In

addition, the pain may worsen with stretching and during exercise (e.g., sprinting, hurdling, and kicking).

Surgical repair of proximal hamstring ruptures also has its inherent risks. Superficial and deep wound infections can occur, as with other surgeries; however, the location of the incision can potentially increase this risk because of the proximity of the incision to the perineum. Additionally, the three main nervous structures at risk of iatrogenic injury are the posterior femoral cutaneous, inferior gluteal, and sciatic nerves. The posterior femoral cutaneous nerve exits the sacral plexus and enters the pelvis through the greater sciatic foramen below the piriformis muscle. It then descends beneath the gluteus maximus with the inferior gluteal artery and runs down the back of the thigh beneath the fascia lata and over the long head of the biceps femoris to the back of the knee.^{60,61} It provides sensation to the skin of the posterior surface of the thigh and leg as well as to the skin of the perineum. It can be injured during the surgical approach for repair if it is not protected. The inferior gluteal nerve is the major innervation of the gluteus maximus, which is the principal extensor of the thigh. This nerve can be injured with aggressive retraction of the gluteus during the surgical approach.⁶¹

The sciatic nerve is the longest and widest single nerve in the human body and provides innervation of the skin of the leg as well as the muscles of the posterior compartment of the thigh; it also provides the motor function of the calf and foot. The nerve is in close proximity to the ischial tuberosity as it runs along the lateral aspect. It may be injured from retraction during exposure of the tuberosity for repair.^{25,29}

Other potential complications associated with proximal hamstring repair include repeat rupture, weakness, and sitting pain. According to the reports on hamstring repair, repeat ruptures are rare. In the cohort in the study by Sarimo et al., 3 of 41 patients failed surgical repair.²⁵ Upon a repeat operation, anatomic repair of the injury could not be achieved. The authors believe that the quality of the tendon can deteriorate with delays in surgical treatment, fatty degeneration, and muscle denervation from nerve injury.²⁵ Several studies have tested postoperative hamstring strength after repair. Recently Carmichael et al.²⁹ found that mean postoperative isotonic strength was 84% compared with the contralateral side; however, other studies have shown a return of strength ranging from 60% to 90% after repair.^{21,26,45,49}

A concern unique to the endoscopic approach is that of fluid extravasation into the pelvis as a result of the fluid used in distending the potential space around the hamstring tendon. Every effort should be made to regularly check the abdomen for any evidence of abdominal distention.

One of the most important aspects in the treatment of proximal hamstring ruptures is early recognition and treatment. Early recognition of the injury allows for early repair of the acute injury, which is substantially easier to perform immediately after the injury occurs (within 4 weeks). Later recognition and delayed surgery lead to a more difficult repair that ultimately may result in increased surgical complications and poorer patient outcomes. Patients who undergo repairs of acute injuries have had better outcomes, as reported in the literature, than those who undergo repairs of chronic injuries.^{24,25}

In summary, recognition of proximal hamstring ruptures allows early treatment with surgical repair. With proper treatment, good functional results can be achieved.

FUTURE CONSIDERATIONS

Few clinical studies have been conducted to test hamstring strength—to assist in determining a range of strength deficit if the tendon is not repaired—in patients who have undergone nonoperative treatment of acute ruptures.²⁴ As a result, in discussing the options of repair or conservative treatment with patients after a diagnosis of a complete proximal hamstring rupture, it is difficult to define the percentage of weakness that may be expected if the repair is not performed.

It will also be interesting to observe if more organized sporting teams begin to implement Nordic exercise programs or others that may be studied in the future to aid in the prevention of hamstring injury.

The emerging field of orthobiologics offers promise of improving outcomes in both the surgical and nonsurgical settings. However, despite its popularity, there is a paucity of high-level evidence to support the use of stem cells or platelet-rich plasma in either the clinical or surgical arena. Targets for research include optimizing injectate composition, stratification of appropriateness based on injury severity and location, and the development of postinjection rehabilitation protocols. Orthobiologics continues to be a popular topic within our societies and their validity should be studied at the highest level.

In addition, further development and refinement of the endoscopic technique is necessary, as at present it is clearly in its earliest phases. The extent to which this technique can be used in all tears remains to be seen. Further studies are necessary to document the outcomes of the technique and compare them with the traditional open procedures.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

Elliott MC, Zarins B, Powell JW, et al. Hamstring muscle strains in professional football players: a 10-year review. *Am J Sports Med*. 2011;39:843–850.

Level of Evidence:

IV

Summary:

The authors of this article provide a summary of the types of strains and ruptures seen in the professional athlete.

Citation:

Miller SL, Gill J, Webb GR. The proximal origin of the hamstrings and surrounding anatomy encountered during repair. A cadaveric study. *J Bone Joint Surg Am*. 2007;89(A):44–48.

Level of Evidence:

Anatomic study

Summary:

The authors of this article provide a review of the pertinent surgical anatomy that delineates the appropriate anatomic areas that must be understood in approaching the area.

Citation:

Sarimo J, Lempainen L, Mattila K, et al. Complete proximal hamstring avulsions: A series of 41 patients with operative treatment. *Am J Sports Med*. 2008;36:1110–1115.

Level of Evidence:

IV

Summary:

This article includes the largest series published, with a good summary of the mechanisms, treatment, and outcomes of the open procedure.

Citation:

Folsom GJ, Larson CM. Surgical treatment of acute versus chronic complete proximal hamstring ruptures: results of a new allograft technique for chronic reconstructions. *Am J Sports Med*. 2008;36:104–109.

Level of Evidence:

II

Summary:

In this article the surgical technique and indications for reconstruction of the hamstring in chronic rupture cases are described.

Citation:

Van der Made AD, Reurink G, Gouttebarga V, et al. Outcome after surgical repair of proximal hamstring avulsions: a systematic review. *Am J Sports Med*. 2015;43(11):2841–2851.

Level of Evidence:

Systematic review

Summary:

Surgical repairs resulted in slightly higher subjective outcomes vs. nonoperative treatments, but there were still deficits compared with baseline and the conclusion was that there was no significant improvement. Delaying repair, both as a primary repair or with an allograft, also showed good outcomes.

REFERENCES

- Brown T. Thigh. In: DeLee JC, Miller MD, eds. *Orthopaedic Sports Medicine: Principles and Practice*. Vol. 2. Philadelphia: Elsevier; 2003.
- Clanton TL. Invited editorial/introduction to nitric oxide and the respiratory musculature: A short history of nitric oxide in skeletal muscle function. Comparative biochemistry and physiology. *Comp Biochem Physiol A Mol Integr Physiol*. 1998;119:165–166.
- Speer KP, Lohnes J, Garrett WE Jr. Radiographic imaging of muscle strain injury. *Am J Sports Med*. 1993;21:89–95.
- Garrett WE Jr, Rich FR, Nikolaou PK, et al. Computed tomography of hamstring muscle strains. *Med Sci Sports Exer*. 1989;21:506–514.
- Culpepper MI, Niemann KM. High school football injuries in Birmingham, Alabama. *South Med J*. 1983;76:873–878.
- Dick R, Ferrara MS, Agel J, et al. Descriptive epidemiology of collegiate men's football injuries: National Collegiate Athletic Association injury surveillance system, 1988–1989 through 2003–2004. *J Athl Train*. 2007;42:221–233.
- Moretz A 3rd, Rashkin A, Grana WA. Oklahoma high school football injury study: a preliminary report. *J Oklahoma State Medical Association*. 1978;71:85–88.
- Powell JW, Barber-Foss KD. Injury patterns in selected high school sports: a review of the 1995–1997 seasons. *J Athl Train*. 1999;34:277–284.
- Shankar PR, Fields SK, Collins CL, et al. Epidemiology of high school and collegiate football injuries in the United States, 2005–2006. *Am J Sports Med*. 2007;35:1295–1303.
- Elliott MC, Zarins B, Powell JW, et al. Hamstring muscle strains in professional football players: a 10-year review. *Am J Sports Med*. 2011;39:843–850.
- Edouard P, Branco P, Alonso J. Muscle injury is the principal injury type and hamstring muscle injury is the first injury diagnosis during top-level international athletics championships between 2007 and 2015. *Br J Sports Med*. 2016;50:619–630.
- Mendiguchia J, Alentorn-Geli E, Brughelli M. Hamstring strain injuries: are we heading in the right direction? *Brit J Sports Med*. 2012;46:81–85.
- Mueller FO, Blyth CS. North Carolina high school football injury study: equipment and prevention. *J Sports Med*. 1974;2:1–10.
- Miller SL, Gill J, Webb GR. The proximal origin of the hamstrings and surrounding anatomy encountered during repair. A cadaveric study. *J Bone Joint Surg*. 2007;89(A):44–48.
- Lyons K, Perry J, Gronley JK, et al. Timing and relative intensity of hip extensor and abductor muscle action during level and stair ambulation: an EMG study. *Phys Ther*. 1983;63:1597–1605.
- Geraci MC Jr, Brown W. Evidence-based treatment of hip and pelvic injuries in runners. *Phys Med Rehabil Clin North Am*. 2005;16:711–747.
- Crema MD, Guermazi A, Tol JL, et al. Acute hamstring injury in football players: association between anatomical location and extent of injury—a large single-center MRI report. *J Sci Med Sport*. 2016;19:317–322.
- Blasier RL, Morawa LG. Complete rupture of the hamstring origin from a water skiing injury. *Am J Sports Med*. 1990;18:435–437.
- Orava S, Kujala UM. Rupture of the ischial origin of the hamstring muscles. *Am J Sports Med*. 1995;23:702–705.
- Sallay PI, Friedman RL, Coogan PG, et al. Hamstring muscle injuries among water skiers. Functional outcome and prevention. *Am J Sports Med*. 1996;24:130–136.
- Klinge KE, Sallay PI. Surgical repair of complete proximal hamstring tendon rupture. *Am J Sports Med*. 2002;30:742–747.
- Mica L, Schwaller A, Stoupis C, et al. Avulsion of the hamstring muscle group: a follow-up of 6 adult non-athletes with early operative treatment: a brief report. *World J Surg*. 2009;33:1605–1610.
- Clanton TO, Coupe KJ. Hamstring strains in athletes: diagnosis and treatment. *J Am Acad Orthop Surg*. 1998;6:237–248.
- Sallay PI, Ballar G, Hamersly S, et al. Subjective and functional outcomes following surgical repair of complete ruptures of the proximal hamstring complex. *Orthopaedics*. 2008;31:1092.
- Sarimo J, Lempainen L, Mattila K, et al. Complete proximal hamstring avulsions: a series of 41 patients with operative treatment. *Am J Sports Med*. 2008;36:1110–1115.
- Brucker PU, Imhoff AB. Functional assessment after acute and chronic complete ruptures of the proximal hamstring tendons. *Knee Surg Sports Traumatol*. 2005;13:411–418.
- Konan S, Haddad F. Successful return to high level sports following early surgical repair of complete tears of the proximal hamstring tendons. *Inter Orthop*. 2010;34:119–123.
- Lempainen L, Sarimo J, Orava S. Recurrent and chronic complete ruptures of the proximal origin of the hamstring muscles repaired with fascia lata autograft augmentation. *Arthroscopy*. 2007;23:441.e1–441.e5.
- Carmichael J, Packam I, Trikha SP, et al. Avulsion of the proximal hamstring origin. Surgical technique. *J Bone Joint Surg Am*. 2008;90A:2365–2374.
- Engelbrechtsen AH, Myklebust G, Holme I, et al. Intrinsic risk factors for hamstring injuries among male soccer players: a prospective cohort study. *Am J Sports Med*. 2010;38:1147–1153.
- Askling CM, Tengvar M, Saartok T, et al. Acute first-time hamstring strains during high-speed running: a longitudinal study including clinical and magnetic resonance imaging findings. *Am J Sports Med*. 2007;35:197–206.
- Comin J, Malliaras P, Baquie P, et al. Return to competitive play after hamstring injuries involving disruption of the central tendon. *Am J Sports Med*. 2013;41:111–115.
- Cacchio A, Borra F, Severini G, et al. Reliability and validity of three pain provocation tests used for the diagnosis of chronic proximal hamstring tendinopathy. *Br J Sports Med*. 2012;46(12):883–887.
- Ekstrand J, Healy JC, Waldén M, et al. Hamstring muscle injuries in professional football: the correlation of MRI findings with return to play. *Br J Sports Med*. 2012;46:112–117.
- Connell DA, Schneider-Kolsky ME, Hoving JL, et al. Longitudinal study comparing sonographic and MRI assessments of acute and healing hamstring injuries. *Am J Roentgenol*. 2004;183:975–984.
- Pettrons P. Ultrasound of muscles. *Eur Radiol*. 2002;12:35–43.
- Ekstrand J, Lee JC, Healy JC. MRI findings and return to play in football: a prospective analysis of 255 hamstring injuries in the UEFA elite club injury study. *Br J Sports Med*. 2016;50:738–743.
- Thompson SM, Fung S, Wood DG. The prevalence of proximal hamstring pathology on MRI in the asymptomatic population. *Knee Surg Sports Traumatol Arthrosc*. 2017;25:108–111.
- De Smet AA, Blankenbaker DG, Alsheik NH, et al. MRI appearance of the proximal hamstring tendons in patients with and without symptomatic proximal hamstring tendinopathy. *Am J Roentgenol*. 2012;198:418–422.

40. Drakos M, Birmingham P, Delos D, et al. Corticosteroid and anesthetic injections for muscle strains and ligament sprains in the NFL. *HSS J*. 2014;10:136–142.
41. Mendiguchia J, Brughelli M. A return-to-sport algorithm for acute hamstring injuries. *Phys Ther Sport*. 2011;12:2–14.
42. Sherry MA, Best TM. J A comparison of 2 rehabilitation programs in the treatment of acute hamstring strains. *Orthop Sports Phys Ther*. 2004;34:116–125.
43. Van der Horst N, Smits D, Peterson J, et al. The preventive effect of the Nordic hamstring exercise on hamstring injuries in amateur soccer players: a randomized controlled trial. *Am J Sports Med*. 2015;43(6):1316–1323.
44. Fader RR, et al. Platelet-rich plasma treatment improves outcomes for chronic proximal hamstring injuries in an athletic population. *Muscles Ligaments Tendons J*. 2014;4(4):461–466.
45. Chakravarthy J, Ramisetty N, Pimpalnerkar A, et al. Surgical repair of complete proximal hamstring tendon ruptures in water skiers and bull riders: a report of four cases and review of the literature. *Br J Sports Med*. 2005;39:569–572.
46. Piposar JR, Amrit V, et al. High-grade partial and retracted (<2 cm) proximal hamstring ruptures: nonsurgical treatment revisited. *Orthop J Sports Med*. 2017;5:1–7.
47. Barnett AJ, Negus JJ, Barton T, et al. Reattachment of the proximal hamstring origin: outcome in patients with partial and complete tears. *Knee Surg Sports Traumatol Arthrosc*. 2015;23:2130–2135.
48. Cohen S, Bradley J. Acute proximal hamstring rupture. *J Am Acad Orthop Surg*. 2007;15:350–355.
49. Cross MJ, et al. Surgical repair of chronic complete hamstring tendon rupture in the adult patient. *Am J Sports Med*. 1998;26:785–788.
50. Guanche CA. Hamstring injuries. *J Hip Preserv Surg*. 2015;2(2):116–122.
51. Kuszewski M, Gnat R, Saulicz E. Stability training of the lumbo-pelvo-hip complex influence stiffness of the hamstrings: a preliminary study. *Scand J Med Sci Sports*. 2009;19:260–266.
52. Folsom GJ, Larson CM. Surgical treatment of acute versus chronic complete proximal hamstring ruptures: results of a new allograft technique for chronic reconstructions. *Am J Sports Med*. 2008;36:104–109.
53. Cohen SR, Vyas A, Bradley DJ. *Functional Results and Outcomes After Repair of Proximal Hamstring Avulsions*. Presented at the AAOS Annual Meeting. San Francisco: 2012.
54. Mansour AA, et al. National football league Athlete's return to play after surgical reattachment of complete proximal hamstring ruptures. *Am J Orthop*. 2013;42(6):E38–E41.
55. Puranen J, Orava S. The hamstring syndrome. A new diagnosis of gluteal sciatic pain. *Am J Sports Med*. 1988;16:517–521.
56. Harris JD, et al. Treatment of proximal hamstring ruptures—a systematic review. *Int J Sports Med*. 2011;32:490–495.
57. Sandmann GH, et al. Mid-term functional outcome and return to sports after proximal hamstring repair. *Int J Sports Med*. 2016;37(7):570–576.
58. Subbu R, et al. Timing of surgery for complete proximal hamstring avulsion injuries: successful clinical outcomes at 6 weeks, 6 months, and after 6 months of injury. *Am J Sports Med*. 2015;43(2):385–391.
59. Van der Made AD, Reurink G, Gouttebauge V, et al. Outcome after surgical repair of proximal hamstring avulsions: A systematic review. *Am J Sports Med*. 2015;43(11):2841–2851.
60. Lempainen L, Sarimo J, Heikkilä J, et al. Surgical treatment of partial tears of the proximal origin of the hamstring muscles. *Br J Sports Med*. 2006;40:688–691.
61. Moore K. *Clinically Oriented Anatomy*. Philadelphia: Williams & Wilkins; 1992.